BANKING LOAN RISK ANALYSIS

API Documentation — Submission Edition

API v1.0.0 | OpenAPI 3.1.0 | Based on supplied FastAPI specification

Includes method, endpoint, request and response format, sample request and sample response for every operation.

Base URL: http://127.0.0.1:8000

Sample payloads below are illustrative, not captured responses from a running server. Sample values must not be treated as verified database records.

1. Authentication and conventions

POST /auth/login is public. Protected routes require the header Authorization: Bearer <JWT_ACCESS_TOKEN>. Role permissions: ADMIN can create, update and delete customers; ADMIN and ANALYST can export CSV; authenticated ADMIN, ANALYST and VIEWER can read customer and analytics data. Role restrictions are based on project implementation, not expressed in the OpenAPI security scheme.

All JSON requests use Content-Type: application/json. Query parameters are URL-encoded. OpenAPI explicitly declares 422 validation errors for selected routes; other runtime errors such as 401, 403, 404 and 500 are not exhaustively declared.

2. Endpoint inventory

Method	Path	Access
GET	/health	Public
GET	/customers	Authenticated
POST	/customers	ADMIN
GET	/customers/{customer_id}	Authenticated
PUT	/customers/{customer_id}	ADMIN
DELETE	/customers/{customer_id}	ADMIN
GET	/customers/{customer_id}/activity	Authenticated
GET	/customers/{customer_id}/risk	Authenticated
GET	/analytics/summary	Authenticated
GET	/analytics/risk-distribution	Authenticated
GET	/analytics/monthly-trends	Authenticated
GET	/analytics/segments	Authenticated
GET	/export/customers	ADMIN/ANALYST
POST	/auth/login	Public
GET	/auth/me	Authenticated
GET	/	Public

3. Endpoint details and examples

3.1 GET /health

Health Check. Check API and database availability.

Authentication: Public.

Parameters: none.

Request format: No request body; no query parameters.

Response format: application/json; declared success status 200.

Sample request:

```
GET /health HTTP/1.1
Host: 127.0.0.1:8000
```

Sample response (illustrative):

```
HTTP/1.1 200 OK
Content-Type: application/json

{
  "status": "healthy",
  "database": "connected"
}
```

Declared response codes: 200 (Successful Response).

3.2 GET /customers

List Customers.

Authentication: Bearer JWT required.

Parameters: page (query, optional, integer); page_size (query, optional, integer); risk_band (query, optional, constrained); has_payment_delay (query, optional, constrained); min_credit_limit (query, optional, constrained); max_credit_limit (query, optional, constrained); sort_by (query, optional, string); sort_order (query, optional, string)

Request format: No request body; query/path parameters as listed above.

Response format: application/json; declared success status 200.

Sample request:

```
GET /customers?page=1&page_size=20 HTTP/1.1
Host: 127.0.0.1:8000
Authorization: Bearer <JWT_ACCESS_TOKEN>
```

Sample response (illustrative):

```
HTTP/1.1 200 OK
Content-Type: application/json

{
  "page": 1,
  "page_size": 20,
  "total_records": 30000,
  "total_pages": 1500,
  "items": [
    {
      "customer_id": 1,
      "age": 24,
      "gender": "Female",
      "education": "University",
      "marital_status": "Married",
      "credit_limit": 20000,
      "has_payment_delay": true,
      "delay_month_count": 2,
      "latest_utilization_ratio": 0.42,
      "risk_score": 4,
      "risk_band": "MEDIUM"
    }
  ]
}
```

Declared response codes: 200 (Successful Response), 422 (Validation Error).

3.3 POST /customers

Create New Customer.

Authentication: Bearer JWT required.

Parameters: none.

Request format: JSON body; no query parameters.

Response format: application/json; declared success status 201.

Sample request:

```
POST /customers HTTP/1.1
Host: 127.0.0.1:8000
Authorization: Bearer <JWT_ACCESS_TOKEN>
Content-Type: application/json

{
  "customer_id": 30001,
  "gender_code": 2,
  "education_code": 2,
  "marital_status_code": 1,
  "age": 30,
  "credit_limit": 50000,
  "default_next_month": 0,
  "monthly_activity": [
    {
      "activity_month": "2005-04-01",
      "repayment_status": 0,
      "bill_amount": 1200.0,
      "payment_amount": 1000.0
    },
    {
      "activity_month": "2005-05-01",
      "repayment_status": 0,
      "bill_amount": 1200.0,
      "payment_amount": 1000.0
    },
    {
      "activity_month": "2005-06-01",
      "repayment_status": 0,
      "bill_amount": 1200.0,
      "payment_amount": 1000.0
    },
    {
      "activity_month": "2005-07-01",
      "repayment_status": 0,
      "bill_amount": 1200.0,
      "payment_amount": 1000.0
    },
    {
      "activity_month": "2005-08-01",
      "repayment_status": 0,
      "bill_amount": 1200.0,
      "payment_amount": 1000.0
    },
    {
      "activity_month": "2005-09-01",
      "repayment_status": 0,
      "bill_amount": 1200.0,
      "payment_amount": 1000.0
    }
  ]
}
```

Sample response (illustrative):

```
HTTP/1.1 201 OK
Content-Type: application/json

{
```

```
  "message": "example",
  "customer_id": 1,
  "risk_score": 1,
  "risk_band": "example"
}
```

Declared response codes: 201 (Successful Response), 422 (Validation Error).

3.4 GET /customers/{customer_id}

Customer Detail.

Authentication: Bearer JWT required.

Parameters: customer_id (path, required, integer)

Request format: No request body; query/path parameters as listed above.

Response format: application/json; declared success status 200.

Sample request:

```
GET /customers/1 HTTP/1.1
Host: 127.0.0.1:8000
Authorization: Bearer <JWT_ACCESS_TOKEN>
```

Sample response (illustrative):

```
HTTP/1.1 200 OK
Content-Type: application/json

{
  "customer_id": 1,
  "demographics": {
    "gender": "example",
    "education": "example",
    "marital_status": "example",
    "age": 1
  },
  "credit_profile": {
    "credit_limit": 100.0,
    "delay_month_count": 1,
    "max_delay_status": 1,
    "has_payment_delay": false,
    "total_bill_amount_6m": 100.0,
    "average_bill_amount_6m": 100.0,
    "total_payment_amount_6m": 100.0,
    "average_payment_amount_6m": 100.0,
    "latest_utilization_ratio": 100.0,
    "default_next_month": 1
  },
  "risk": {
    "payment_to_bill_ratio": 100.0,
    "delay_frequency_score": 1,
    "delay_severity_score": 1,
    "utilization_score": 1,
    "payment_behavior_score": 1,
    "risk_score": 1,
    "risk_band": "example",
    "rule_version": "example"
  }
}
```

Declared response codes: 200 (Successful Response), 422 (Validation Error).

3.5 PUT /customers/{customer_id}

Update Existing Customer.

Authentication: Bearer JWT required.

Parameters: customer_id (path, required, integer)

Request format: JSON body; query/path parameters as listed above.

Response format: application/json; declared success status 200.

Sample request:

```
PUT /customers/1 HTTP/1.1
Host: 127.0.0.1:8000
Authorization: Bearer <JWT_ACCESS_TOKEN>
Content-Type: application/json

{
  "gender_code": 2,
  "education_code": 2,
  "marital_status_code": 1,
  "age": 31,
  "credit_limit": 55000,
  "default_next_month": 0,
  "monthly_activity": [
    {
      "activity_month": "2005-04-01",
      "repayment_status": 0,
      "bill_amount": 1200.0,
      "payment_amount": 1000.0
    },
    {
      "activity_month": "2005-05-01",
      "repayment_status": 0,
      "bill_amount": 1200.0,
      "payment_amount": 1000.0
    },
    {
      "activity_month": "2005-06-01",
      "repayment_status": 0,
      "bill_amount": 1200.0,
      "payment_amount": 1000.0
    },
    {
      "activity_month": "2005-07-01",
      "repayment_status": 0,
      "bill_amount": 1200.0,
      "payment_amount": 1000.0
    },
    {
      "activity_month": "2005-08-01",
      "repayment_status": 0,
      "bill_amount": 1200.0,
      "payment_amount": 1000.0
    },
    {
      "activity_month": "2005-09-01",
      "repayment_status": 0,
      "bill_amount": 1200.0,
      "payment_amount": 1000.0
    }
  ]
}
```

Sample response (illustrative):

```
HTTP/1.1 200 OK
Content-Type: application/json

{
  "message": "example",
  "customer_id": 1,
  "risk_score": 1,
  "risk_band": "example"
}
```

Declared response codes: 200 (Successful Response), 422 (Validation Error).

3.6 DELETE /customers/{customer_id}

Delete Existing Customer.

Authentication: Bearer JWT required.

Parameters: customer_id (path, required, integer)

Request format: No request body; query/path parameters as listed above.

Response format: application/json; declared success status 200.

Sample request:

```
DELETE /customers/1 HTTP/1.1
Host: 127.0.0.1:8000
Authorization: Bearer <JWT_ACCESS_TOKEN>
```

Sample response (illustrative):

```
HTTP/1.1 200 OK
Content-Type: application/json

{
  "message": "example",
  "customer_id": 1
}
```

Declared response codes: 200 (Successful Response), 422 (Validation Error).

3.7 GET /customers/{customer_id}/activity

Customer Activity.

Authentication: Bearer JWT required.

Parameters: customer_id (path, required, integer)

Request format: No request body; query/path parameters as listed above.

Response format: application/json; declared success status 200.

Sample request:

```
GET /customers/1/activity HTTP/1.1
Host: 127.0.0.1:8000
Authorization: Bearer <JWT_ACCESS_TOKEN>
```

Sample response (illustrative):

```
HTTP/1.1 200 OK
Content-Type: application/json

{
  "customer_id": 1,
  "activity": [
    {
      "activity_month": "2005-09-01",
      "repayment_status": 1,
      "bill_amount": 100.0,
      "payment_amount": 100.0
    }
  ]
}
```

Declared response codes: 200 (Successful Response), 422 (Validation Error).

3.8 GET /customers/{customer_id}/risk

Customer Risk.

Authentication: Bearer JWT required.

Parameters: customer_id (path, required, integer)

Request format: No request body; query/path parameters as listed above.

Response format: application/json; declared success status 200.

Sample request:

```
GET /customers/1/risk HTTP/1.1
Host: 127.0.0.1:8000
Authorization: Bearer <JWT_ACCESS_TOKEN>
```

Sample response (illustrative):

```
HTTP/1.1 200 OK
Content-Type: application/json

{
  "customer_id": 1,
  "payment_to_bill_ratio": 0.65,
  "delay_frequency_score": 1,
  "delay_severity_score": 1,
  "utilization_score": 1,
  "payment_behavior_score": 1,
  "risk_score": 4,
  "risk_band": "MEDIUM",
  "rule_version": "v1"
}
```

Declared response codes: 200 (Successful Response), 422 (Validation Error).

3.9 GET /analytics/summary

Portfolio Summary.

Authentication: Bearer JWT required.

Parameters: none.

Request format: No request body; no query parameters.

Response format: application/json; declared success status 200.

Sample request:

```
GET /analytics/summary HTTP/1.1
Host: 127.0.0.1:8000
Authorization: Bearer <JWT_ACCESS_TOKEN>
```

Sample response (illustrative):

```
HTTP/1.1 200 OK
Content-Type: application/json

{
  "total_customers": 1,
  "total_credit_exposure": 100.0,
  "average_credit_limit": 100.0,
  "historical_defaults": 1,
  "historical_default_rate_pct": 100.0,
  "customers_with_payment_delay": 1,
  "customers_with_repeated_delay": 1,
  "average_latest_utilization_pct": 100.0,
  "over_limit_customers": 1,
  "total_bill_amount_6m": 100.0,
  "total_payment_amount_6m": 100.0
}
```

Declared response codes: 200 (Successful Response).

3.10 GET /analytics/risk-distribution

Risk Distribution.

Authentication: Bearer JWT required.

Parameters: none.

Request format: No request body; no query parameters.

Response format: application/json; declared success status 200.

Sample request:

```
GET /analytics/risk-distribution HTTP/1.1
Host: 127.0.0.1:8000
Authorization: Bearer <JWT_ACCESS_TOKEN>
```

Sample response (illustrative):

```
HTTP/1.1 200 OK
Content-Type: application/json

[
  {
    "risk_band": "LOW",
    "total_customers": 13966,
    "historical_defaults": 1000,
    "historical_default_rate_pct": 7.16,
    "average_risk_score": 1.3
  }
]
```

Declared response codes: 200 (Successful Response).

3.11 GET /analytics/monthly-trends

Monthly Trends.

Authentication: Bearer JWT required.

Parameters: none.

Request format: No request body; no query parameters.

Response format: application/json; declared success status 200.

Sample request:

```
GET /analytics/monthly-trends HTTP/1.1
Host: 127.0.0.1:8000
Authorization: Bearer <JWT_ACCESS_TOKEN>
```

Sample response (illustrative):

```
HTTP/1.1 200 OK
Content-Type: application/json

[
  {
    "activity_month": "2005-09-01",
    "customer_records": 1,
    "total_bill_amount": 100.0,
    "average_bill_amount": 100.0,
    "total_payment_amount": 100.0,
    "average_payment_amount": 100.0,
    "delayed_customers": 1,
    "delayed_customer_rate_pct": 100.0
  }
]
```

Declared response codes: 200 (Successful Response).

3.12 GET /analytics/segments

Segment Analytics.

Authentication: Bearer JWT required.

Parameters: segment (query, required, string)

Request format: No request body; query/path parameters as listed above.

Response format: application/json; declared success status 200.

Sample request:

```
GET /analytics/segments?segment=gender HTTP/1.1
Host: 127.0.0.1:8000
Authorization: Bearer <JWT_ACCESS_TOKEN>
```

Sample response (illustrative):

```
HTTP/1.1 200 OK
Content-Type: application/json

[
  {
    "segment": "example",
    "total_customers": 1,
    "average_credit_limit": 100.0,
    "historical_defaults": 1,
    "historical_default_rate_pct": 100.0,
    "customers_with_payment_delay": 1,
    "average_latest_utilization_pct": 100.0
  }
]
```

Declared response codes: 200 (Successful Response), 422 (Validation Error).

3.13 GET /export/customers

Export Customers.

Authentication: Bearer JWT required.

Parameters: risk_band (query, optional, constrained); has_payment_delay (query, optional, constrained); min_credit_limit (query, optional, constrained); max_credit_limit (query, optional, constrained); sort_by (query, optional, string); sort_order (query, optional, string)

Request format: No request body; query/path parameters as listed above.

Response format: text/csv attachment (runtime behavior; OpenAPI response schema unspecified); declared success status 200.

Sample request:

```
GET /export/customers?risk_band=LOW&has_payment_delay=false HTTP/1.1
Host: 127.0.0.1:8000
Authorization: Bearer <JWT_ACCESS_TOKEN>
```

Sample response (illustrative):

```
HTTP/1.1 200 OK
Content-Type: text/csv

customer_id,credit_limit,risk_score,risk_band
1,20000,2,LOW
```

Declared response codes: 200 (Successful Response), 422 (Validation Error).

3.14 POST /auth/login

Login.

Authentication: Public.

Parameters: none.

Request format: JSON body; no query parameters.

Response format: application/json; declared success status 200.

Sample request:

```
POST /auth/login HTTP/1.1
Host: 127.0.0.1:8000
Content-Type: application/json

{
  "username": "demo_admin",
  "password": "<PASSWORD>"
}
```

Sample response (illustrative):

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
{
  "access_token": "<JWT_ACCESS_TOKEN>",
  "token_type": "bearer",
  "expires_in": 1800,
  "user_id": 1,
  "username": "demo_admin",
  "role": "ADMIN"
}
```

Declared response codes: 200 (Successful Response), 422 (Validation Error).

3.15 GET /auth/me

Get Authenticated User. Return the currently authenticated user's profile.

Authentication: Bearer JWT required.

Parameters: none.

Request format: No request body; no query parameters.

Response format: application/json; declared success status 200.

Sample request:

```
GET /auth/me HTTP/1.1
Host: 127.0.0.1:8000
Authorization: Bearer <JWT_ACCESS_TOKEN>
```

Sample response (illustrative):

```
HTTP/1.1 200 OK
Content-Type: application/json

{
  "user_id": 1,
  "username": "demo_admin",
  "email": "admin@example.com",
  "role": "ADMIN"
}
```

Declared response codes: 200 (Successful Response).

3.16 GET /

Root.

Authentication: Public.

Parameters: none.

Request format: No request body; no query parameters.

Response format: application/json; declared success status 200.

Sample request:

```
GET / HTTP/1.1
Host: 127.0.0.1:8000
```

Sample response (illustrative):

```
HTTP/1.1 200 OK
Content-Type: application/json

{
  "message": "Banking Loan Risk Analysis API"
}
```

Declared response codes: 200 (Successful Response).

4. Reference and accuracy notes

Source of endpoint definitions: supplied FastAPI OpenAPI 3.1 JSON (Pasted markdown(4).md). 13 unique paths, 16 HTTP operations. The supplied specification does not define a CSV response schema for GET /export/customers or response schemas for GET /health and GET /. Their examples are indicative only. Role mapping comes from project implementation context. The six-month activity examples demonstrate schema shape, not real customer records.

Local interactive documentation: http://127.0.0.1:8000/docs | OpenAPI: http://127.0.0.1:8000/openapi.json